

DESIGN AND IMPLEMENTATION OF A SOC MONITORING AND THREAT INVESTIGATION SYSTEM USING SPLUNK ENTERPRISE

Prepared By:
Ademuyiwa Yinus

Project Type:
SOC Monitoring Threat Investigation System Using Splunk Enterprise

Date:
June 2026

INTRODUCTION

Security Operation Centers (SOCs) play a critical role in modern cybersecurity by continuously monitoring, detecting, investigating and responding to security threats. As organizations generate large volumes of security logs daily, Security Information and Event Management (SIEM) platforms such as Splunk Enterprise help security analysts centralize and analyze security events for effective threat detection.

This project focuses on the design and implementation of a SOC Monitoring and Threat Investigation System using Splunk Enterprise. The objective was to collect, monitor, analyze and visualize security events from the MidnightSwap dataset to identify malicious activities, investigate attack patterns, and provide actionable security insights through an interactive dashboard.

PROJECT OBJECTIVES

The objectives of this project were:

- To deploy and configure Splunk Enterprise as a SIEM platform.
- To Ingest and analyze security logs.
- To identify malicious activities and attack indicators.
- To investigate suspicious IP addresses and targeted user accounts.
- To visualize security events through a SOC dashboard.
- To improve threat detection and incident investigation capabilities.

TOOLS AND TECHNOLOGY USED

The following tools were used during the project:

- Splunk Enterprise
- Kali Linux
- Oracle VirtualBox
- Search Processing Language(SPL)
- MidnightSwap Security Dataset

SYSTEM ARCHITECTURE

The SOC Monitoring System was implemented using Splunk Enterprise running on a Kali Linux virtual machine.

The workflow included:

1. Security logs were ingested into Splunk.
2. Splunk indexed the security events.
3. SPL queries were used to analyze attack patterns.
4. Dashboard visualizations were created to support threat investigation.
5. Security findings were generated based on detected attack activities.

DATA ANALYSIS AND INVESTIGATION

The dataset contained various attack indicators and suspicious security events.

Several SPL queries were developed to identify:

Total Security Events

The system processed a total of 17,169 security events, this metric provides visibility into the overall volume of monitored activities.

Unique Devices

A total of 3,061 unique devices were identified across the dataset.

Attack Flags Detected

The investigation identified 19 attack categories, indicating multiple threat activities within the environment.

DASHBOARD DEVELOPMENT

A comprehensive SOC Monitoring and Threat Investigation Dashboard was developed to visualize security events.

The dashboard includes:

Attack Distribution by Flag

This visualization displays the distribution of attack categories detected within the environment.

Key attack categories observed include:

- BACKGROUND_BRUTE_FORCE
- PHISHING_ACCOUNT_TAKEOVER
- PHISHING_CAMPAGN_NOV14
- PHISHING_MASS_PROBE
- RANSOMWARE_C2
- RANSOMWARE_IMPACT

The analysis showed that BACKGROUND_BRUTE_FORCE represented the highest volume of attack activity. Additional investigation of the dataset revealed occurrences of the MIDNIGHT_SWAP_ATTACK flag, with approximately 25 events recorded. Although it was not

among the most dominant attack categories displayed in the dashboard visualization, it represented notable suspicious activity within the environment and was included in the overall threat analysis.

Top Attack Types

A bar chart was developed to identify the most common attack categories. Findings showed that brute-force attacks were significantly higher than other attack types.

Top 10 Attacking IP Addresses

This panel identifies the IP addresses generating the highest number of suspicious activities. To maintain privacy and demonstrate security best practices, IP addresses were masked before presentation on the dashboard.

Top Targeted Usernames

This panel identifies user account most frequently targeted by malicious activities. The analysis revealed that several user accounts experienced repeated attack attempts, indicating possible credential-focused attacks.

Attack Timeline

A timeline visualization was developed to display attack trends over time. This allowed identification of attack spikes and periods of increased malicious activity.

KEY FINDINGS

The investigation revealed the following:

- Brute-force attacks were the most dominant attack type.
- Multiple phishing-related activities were detected.
- Several IP addresses repeatedly generated suspicious events.
- Certain user accounts were targeted more frequently than others.
- Attack activity varied over time with noticeable spikes indicating concentrated attack campaigns.

CHALLENGES ENCOUNTERED

During the project, several challenges were encountered:

- Data normalization and field extraction.
- Dashboard design optimization.
- SPL query development and troubleshooting.
- Resource limitations within the virtualized environment.
- IP masking implementation for secure reporting.

These challenges were successfully resolved through query refinement and system optimization.

CONCLUSION

This project successfully demonstrated the design and implementation of a SOC Monitoring and Threat Investigation System using Splunk Enterprise.

The dashboard provided centralized visibility into security events, attack trends, malicious IP activity, targeted users, and threat categories. The project highlights the effectiveness of SIEM solutions in supporting security monitoring, threat detection, and incident investigation activities.

The implementation strengthened practical skills in Splunk administration, security monitoring, log analysis, dashboard development, and SOC operations.

RECOMMENDATIONS

Based on the findings, the following recommendations are proposed:

- Implement stronger account lockout policies to reduce brute-force attacks.
- Deploy multi-factor authentication (MFA).
- Continuously monitor high-risk IP addresses.
- Conduct regular phishing awareness training.
- Implement automated alerting for critical attack indicators.
- Enhance SOC monitoring through threat intelligence integration.